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Small Sounding Rocket Boost Phase Roll Rate

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Introduction

Roll rate is an important parameter for both sounding rocket dynamics and, in some cases, payload operations. And, it often appears easy to analyze because, as long as the angle of attack is small, the governing equation is both linear and decoupled from other rocket dynamics. Thus, given a solution for the trajectory, a simple numerical integration will provide the roll history. However, here we eek even simpler descriptions of the roll behavior.

Much of this already exists. Reference 2 tells most of what one would like to know about the roll rate after burnout. As an incidental analysis, ref. (1) provides a partial description of boost phase roll motion. This memo documents the derivation of the boost phase roll motion used in ref. (1).

Mnemonics

Mnemonic	Defintion
a	Acceleration along the flight path, ft/sec^2
C_L	Roll moment coefficient = Roll moment / qSd
$C_{L\delta}$	$\frac{\partial C_L}{\partial \delta}$ for a single fin panel
C_{Lp}	$\frac{\partial C_L}{\partial (pd/2U)}$ for the entire rocket
d	Aerodynamic reference length, ft $=2R$
h	Altitude, ft
I_R	Roll moment of inertia, sl-ft^2
L	Launcher length, ft
N	Number of fin panels
p	Roll rate, rad/sec
q	Dynamic pressure, lb/ft^2 $=\rho U^2/2$
R	Body radius, ft
S	Aerodynamic reference area, ft^2 $=\pi R^2$
U	Flight speed, ft/sec
δ_i	Cant angle of the ith fin panel, rad
λ_R	Roll wave number, rad/ft $=p/U$
ϕ	Roll angle change after separation from the launcher, rad
ρ	Atmospheric mass density, sl/ft^3
$(.)_{IG}$	Value of $(.)$ at ignition of an upper stage

Elementary Analysis

Begin by writing the roll dynamical equation of motion, assuming a purely vertical flight:

$$I_R \frac{dp}{dt} = qSd\left[C_{Lp}(\frac{pd}{2U}) + C_{L\delta}\sum_i \delta_i \right] \tag{1}$$

During most of the boost phase a "steady state" solution can be easily found by neglecting the roll moment of inertia term:

$$p \cong - \frac{2UC_{L\delta}}{dC_{Lp}} \sum_i \delta_i \tag{2a}$$

This can be rewritten in terms of the roll wave number:

$$\lambda_R = - \frac{2C_{L\delta}}{dC_{Lp}} \sum_i \delta_i \, . \tag{2b}$$

Equation (2b) is a very important result. Apart from minor variations due to Mach number and aeroelasticity, it shows that λ_R will remain constant during boost. Therefore, during boost,

$$p = \lambda_R U \tag{2c}$$

It follows that roll rate is proportional to flight speed.

The roll angle estimation follows similar lines.

$$\frac{d\phi}{dt} = p = U \frac{d\phi}{dh} \tag{3a}$$

Substituting from eq. (2c) into eq. (3a) results in

$$\frac{d\phi}{dh} = \lambda_R \text{, or} \tag{3b}$$

$$\Delta\phi = \lambda_R (h - L) \, . \tag{3c}$$

Compare eq's. (2c) and (3c). The similarity is striking.

Corrections to the Elementary Theory

The roll moment of inertia term should not be always and entirely neglected. For example, during "steady state" boost the driving torques from fin cant must not only overcome the damping torque but also add to the roll angular momentum. Most importantly, there will be a significant transient after ignition when the roll inertia dominates the initial response. Keep in mind that the roll rate is zero until the launcher is cleared.

It's convenient to change notation at this point. Let λ_{R0} be the first approximation as given by eq. (2b). This is the asymptotic result for large dynamic pressure. Now return to eq. (1), and rewrite it as

$$\frac{I_R a}{qSd}\frac{dp}{dU} = C_{Lp}\frac{pR}{U} + C_{L\delta}\sum_i \delta_i \,. \tag{4a}$$

Now, if eq. (2c) is used, we find that

$$\frac{I_R a \lambda_{R0}}{qSdRC_{Lp}} + \lambda_{R0} = \frac{p}{U} = \lambda_{R1}, \tag{4b}$$

where λ_{R1} is an improved approximation. The dimensionless parameter,

$$J = \frac{I_R a}{qSdRC_{Lp}}, \tag{4c}$$

governs how much the roll rate lags its elementary value as given by eq. (2c). Using the constant acceleration approximation,

$$U^2 = 2ah, \tag{4d}$$

J can be written as

$$J = \frac{I_R}{2\pi\rho h R^4 C_{Lp}}. \tag{4e}$$

That is,

$$\lambda_{R1} = (1 + J(h))\lambda_{R0}. \tag{4f}$$

is an improved estimate on the "steady state" roll rate. Note that J will always be negative because C_{Lp} is negative. Thus, the improved roll rate first approximation will always be smaller than its asymptotic value, especially at lower altitudes.

The same adjustment can be made to the roll angle change. Substitute eq's. (4c) and (4f) into eq. (3b):

$$\Delta\phi = \lambda_{R0}\left[h - L + \frac{I_R}{2\pi\rho R^4 C_{Lp}}\log\left(\frac{h}{L}\right)\right]. \tag{4g}$$

Once Again, with Passion

While it's apparent that eq. (4f) gives an improved approximation, it is only the leading term in an asymptotic expansion for the roll rate. If the process leading to eq. (4b) is used repetitively, the full asymptotic expansion is found to be

$$\frac{\lambda_{RN}}{\lambda_{R0}} = 1 + J + J^2 + \ldots + J^N . \tag{5a}$$

This series can be summed to give

$$\frac{\lambda_{R\infty}}{\lambda_{R0}} = \frac{1}{1 - J} . \tag{5b}$$

Equation (5b) matches the zero roll rate initial condition only when the launcher length L is zero. An empirical adjustment that causes the initial condition to be satisfied for any launcher length is

$$J \cong \frac{I_R}{2\pi\rho(h - L)R^4 C_{Lp}} . \tag{5c}$$

Equation (5b) using eq. (5c) is plotted below together with eq. (2b) and the flight test data for Prospector -18D. This rocket does not leave its launcher until TALO ≈ 1 sec. Up to about TALO = 6 sec, it can be seen that the agreement is satisfactory. The "hockey stick" upturn after 6 sec is thought to be caused by fin static aeroelastic effects.

The integral for the roll angle change is straightforward:

$$\Delta\phi = \lambda_{R0}\left[h - L + k \log\frac{h - L - k}{-k}\right], \tag{5d}$$

where

$$k = \frac{I_R}{2\pi\rho R^4 C_{Lp}} . \tag{5e}$$

More generally, consider the conditions at the ignition of an upper stage. We need to match a given initial roll wave number at a specified altitude. Assuming the upper stage were separated long before its ignition it remains to select an artificial altitude h^* that will cause the roll rate to match its initial value:

$$\lambda_R = \frac{\lambda_{R0}}{1 - 2\pi\rho(h - h^*)R^4 C_{Lp}} .$$

(5f)

At ignition,

$$h_{IG} - h^* = \frac{\frac{\lambda_{R0}}{\lambda_{RIG}} - 1}{-2\pi\rho R^4 C_{Lp}} . \tag{5g}$$

Now, if the upper stage were separated well before its ignition,

$$\lambda_{R0} \cong \lambda_{RIG},$$

and hence $h^* \cong h_{IG}$.

Thus,
$$\lambda_R \cong \frac{\lambda_{R0}}{1 - 2\pi\rho(h - h_{IG})R^4 C_{Lp}},$$

(5h)

and
$$\Delta\phi \cong \lambda_{R0}\left[h - h_{IG} + k\log\frac{h - h_{IG} - k}{-k}\right]. \qquad (5i)$$

References

1. "Small Sounding Rocket Apogee Roll Rate", by C. P. Hoult, RST memo, 15 January 2013
2. "Sounding Rocket Structural Design Loads (rev.7)", by C. P. Hoult, RST memo, 24 March 2013

